

Data Communication

Evaluation

Final Exam	Mid	Work year	Lab	Total
60	20	10	10	100

Chapter 1

Introduction

Chapter 1

In Chapter 1, we introduce the concepts of data communications and networking. We discuss data communications components, data representation, and data flow. We then move to the structure of networks that carry data. We discuss network topologies, categories of networks, and the general idea behind the Internet. The section on protocols and standards gives a quick overview of the organizations that set standards in data communications and networking.

Chapter 2

The two dominant networking models are the Open Systems Interconnection (OSI) and the Internet model (TCP/IP). The first is a theoretical framework; the second is the actual model used in today's data communications. In Chapter 2, we first discuss the OSI model to give a general background. We then concentrate on the Internet model, which is the foundation for the rest of the book.

Chapter 3

Chapter 3 discusses the relationship between data, which are created by a device, and electromagnetic signals, which are transmitted over a medium.

Chapter 4

Chapter 4 deals with digital transmission. We discuss how we can convert digital or analog data to digital signals.

Chapter 5

Chapter 5 deals with analog transmission. We discuss how we can convert digital or analog data to analog signals.

Chapter 6

Chapter 6 shows how we can use the available bandwidth efficiently. We discuss two separate, but related topics, multiplexing and spreading.

1-1 DATA COMMUNICATIONS

*The term **telecommunication** means communication at a distance. The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data. **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.*

Topics discussed in this section:

- **Components of a data communications system**
- **Data Flow**

Data Representation

- Text: represented using bit patterns
 - ASCII: 7-bit (128 symbols)
 - Extended ASCII: 8-bit (256 symbols)
 - Unicode: 16-bit (65,536 symbols)
 - ISO: 32-bit (4B symbols)
- Numbers: represented in binary
- Images: bit patterns for pixels and colors
- Audio: analog or digitized form
- Video: analog or digitized form

Figure 1.1 *Components of a data communication system*

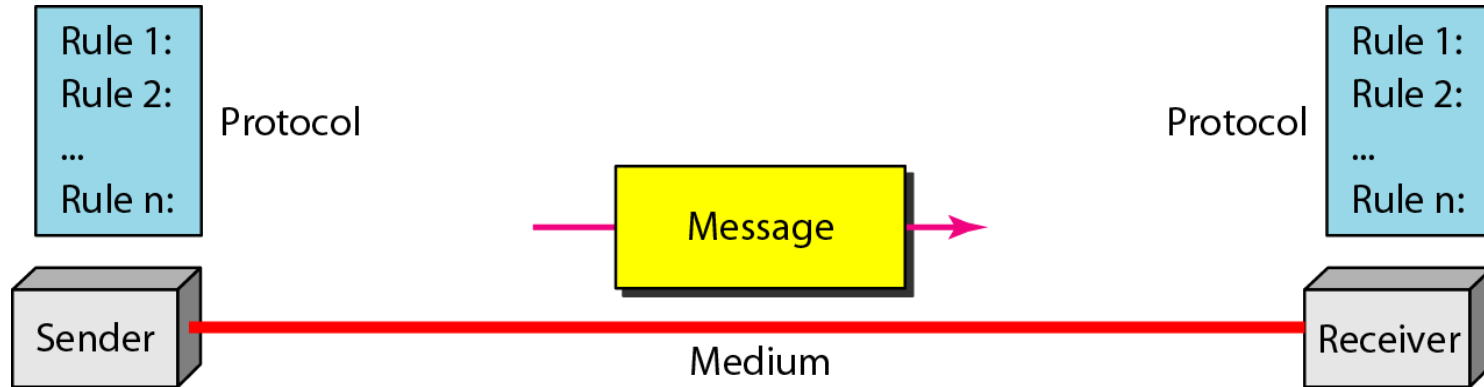
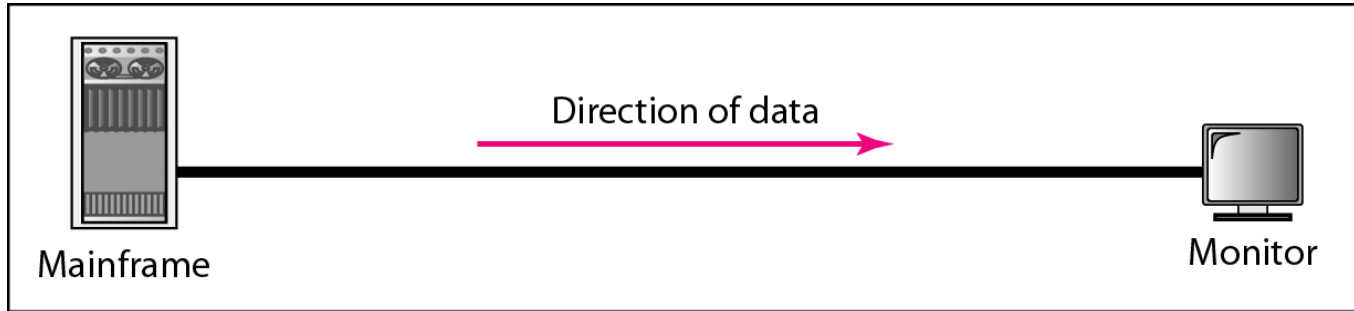
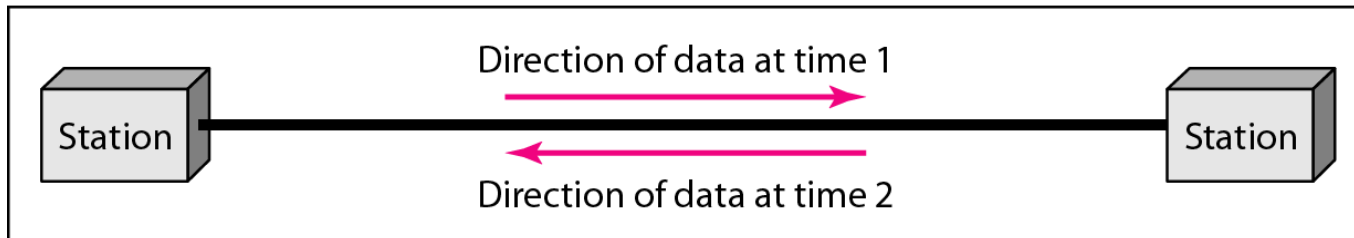


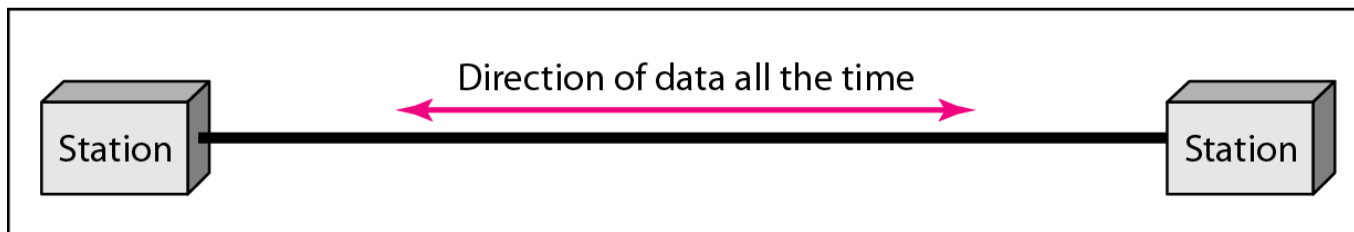
Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



a. Simplex



b. Half-duplex



c. Full-duplex

1-2 NETWORKS

*A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network. A link can be a cable, air, optical fiber, or any medium which can transport a signal carrying information.*

Topics discussed in this section:

- Network Criteria
- Physical Structures
- Categories of Networks

Network Criteria

- **Performance**

- Depends on Network Elements
- Measured in terms of Delay and Throughput

- **Reliability**

- Failure rate of network components
- Measured in terms of availability/robustness

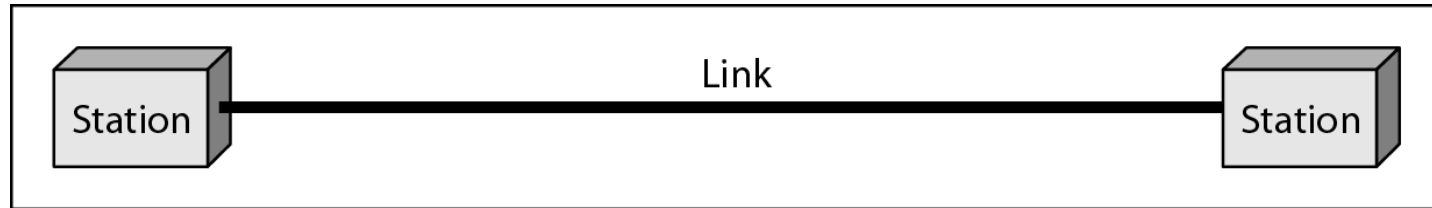
- **Security**

- Data protection against corruption/loss of data due to:
 - Errors
 - Malicious users

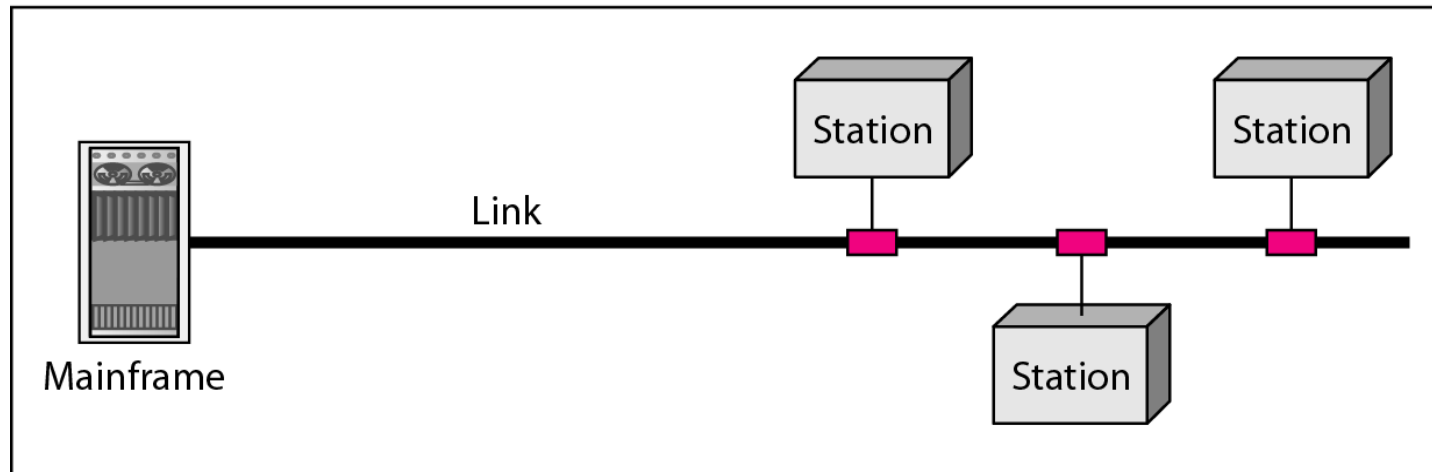
Physical Structures

- **Type of Connection**
 - Point to Point - single transmitter and receiver
 - Multipoint - multiple recipients of single transmission
- **Physical Topology**
 - Connection of devices
 - Type of transmission - unicast, mulitcast, broadcast

Figure 1.3 *Types of connections: point-to-point and multipoint*



a. Point-to-point



b. Multipoint

-Type of transmission - unicast, mulitcast, broadcast

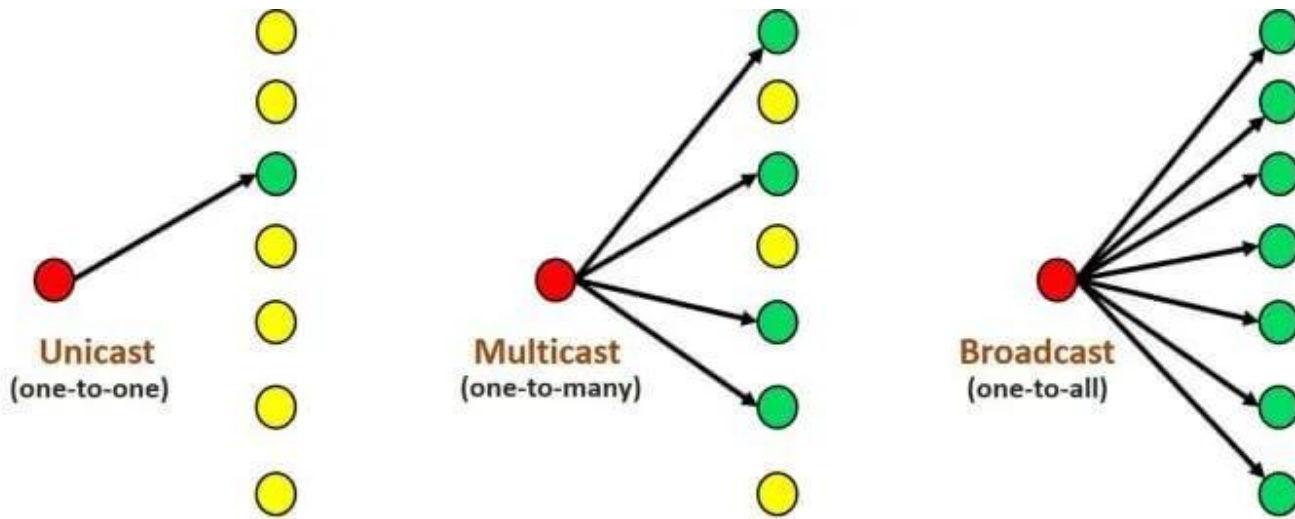


Figure 1.4.1 Type of transmission - unicast, mulitcast, broadcast

Physical Topology

Connection of devices

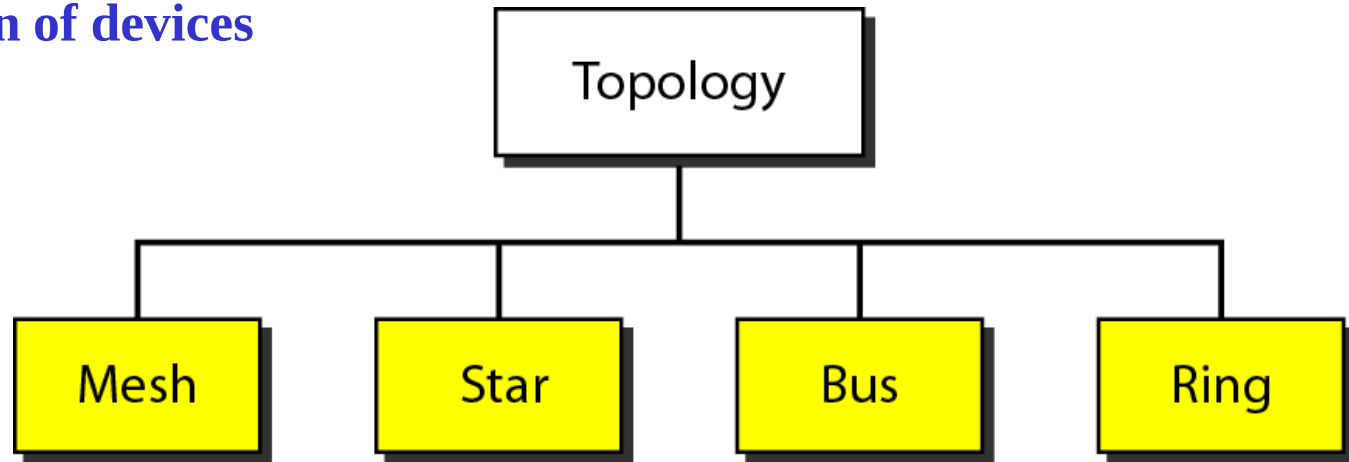
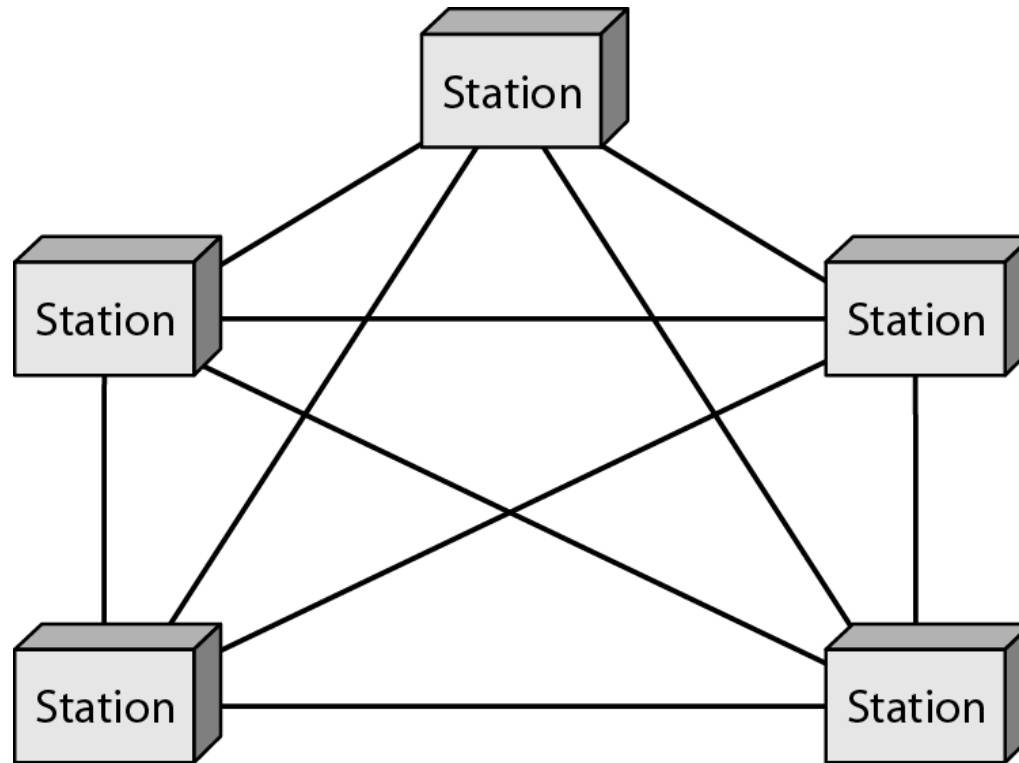


Figure 1.4 *Categories of topology*

Physical topology describes a network's actual or physical layout, i.e., wires, media (computers), or cables. Various types of physical topology include star, bus, mesh, and ring topology

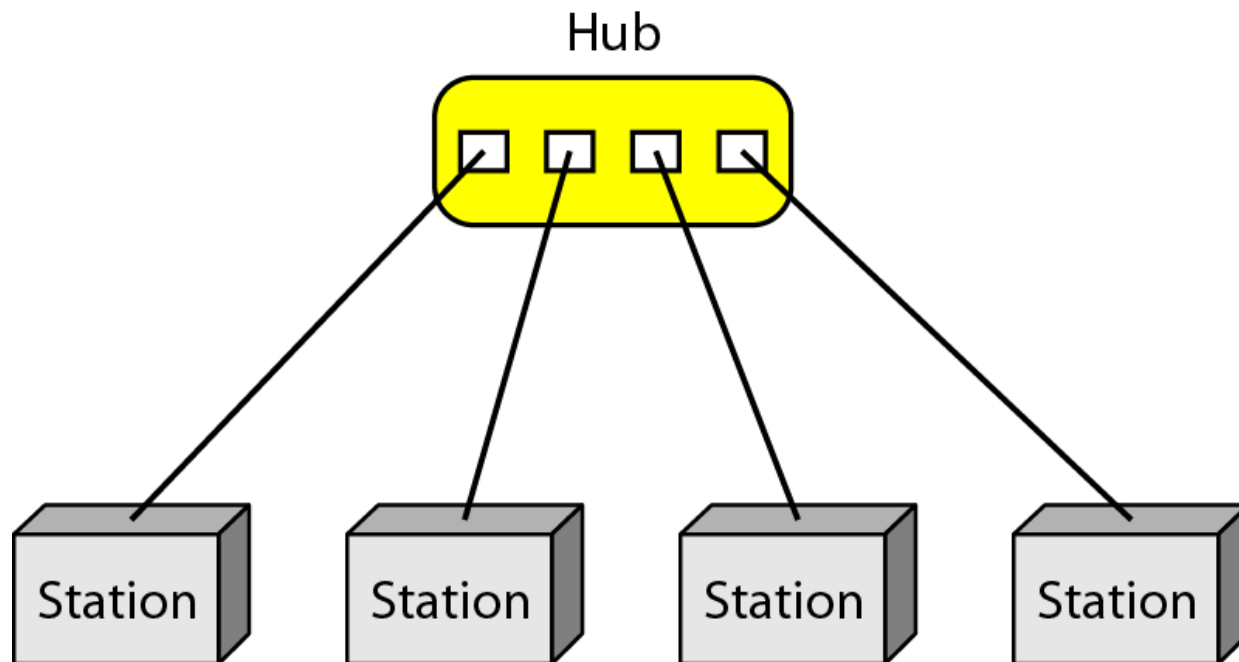
Figure 1.5 *A fully connected mesh topology (five devices)*



Mesh Topology

- Point-to-point
- $n(n-1)/2$ dedicated links
- $(n-1)$ I/O ports per device
- **Advantages:**
 - Full capacity
 - Robustness (one link failure does not fail whole network)
 - Privacy and security
 - Easy fault detection and isolation
- **Disadvantages:**
 - Cabling (installation)
 - I/O ports
 - Space
 - Cost

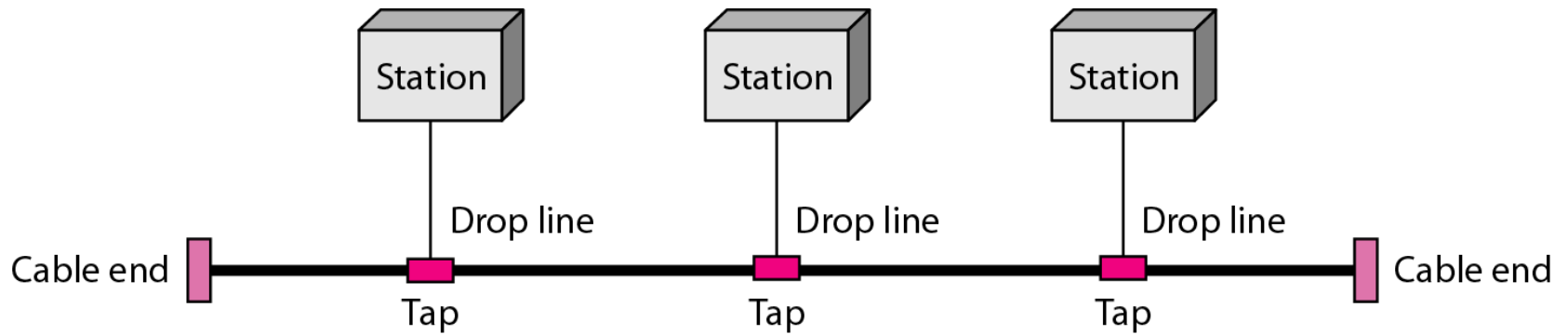
Figure 1.6 *A star topology connecting four stations*



Star Topology

- N dedicated point-to-point links to the hub
- Indirect connections among the nodes
- 1 I/O port per device
- **Advantages:**
 - Robustness
 - Easy fault detection and isolation
- **Disadvantages:**
 - Single point of failure at the hub
 - Bottleneck at the hub.

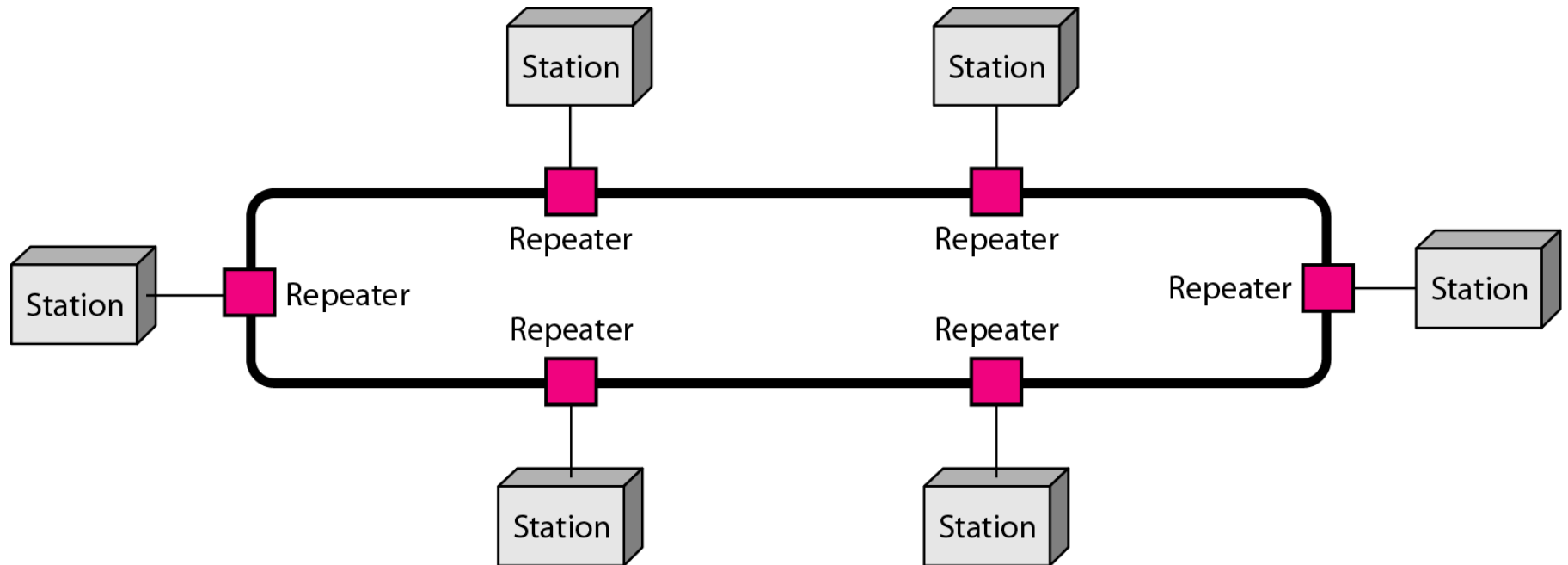
Figure 1.7 *A bus topology connecting three stations*



Bus Topology

- Multi-point
- 1 backbone link
- 1 I/O port per device
- **Advantages:**
 - Ease of installation
 - Minimum cabling at installation
- **Disadvantages:**
 - Limited taps and distance to node
 - Limited node addition after initial installation
 - Limited quality due to signal reflection at taps
 - Difficult fault isolation

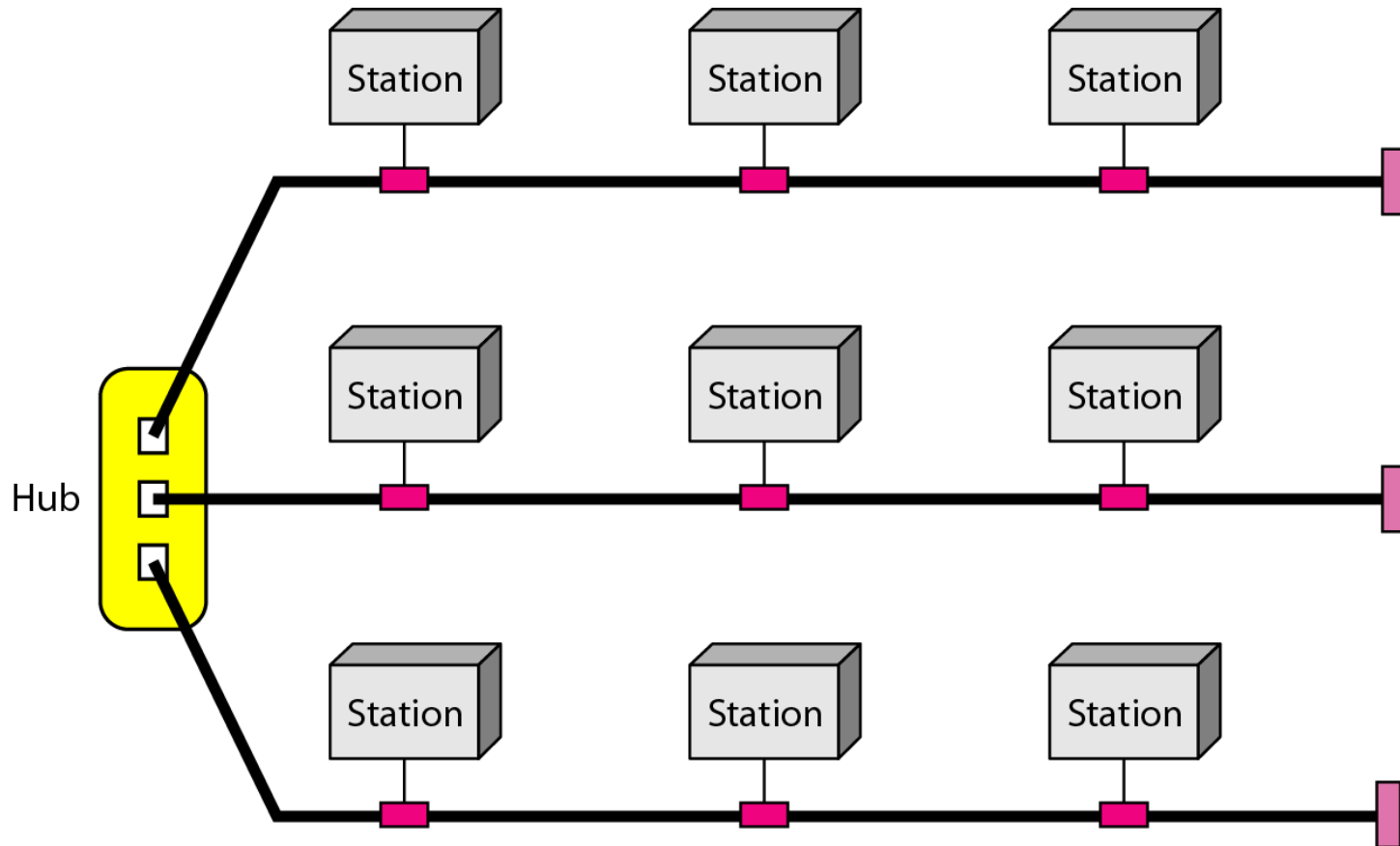
Figure 1.8 *A ring topology connecting six stations*



Ring Topology

- Point-to-point
- n dedicated links
- 2 I/O ports per device
- **Advantages:**
 - Easy installation
 - Easy fault detection and isolation
- **Disadvantages:**
 - Speed: store-and-forward at each node
 - Ring length and number of nodes

Figure 1.9 *A hybrid topology: a star backbone with three bus networks*



Categories of Networks

- **Local Area Networks (LANs)**
 - Short distances
 - Designed to provide local interconnectivity
- **Wide Area Networks (WANs)**
 - Long distances
 - Provide connectivity over large areas
- **Metropolitan Area Networks (MANs)**
 - Provide connectivity over areas such as a city, a campus

Figure 1.10 *An isolated LAN connecting 12 computers to a hub in a closet*

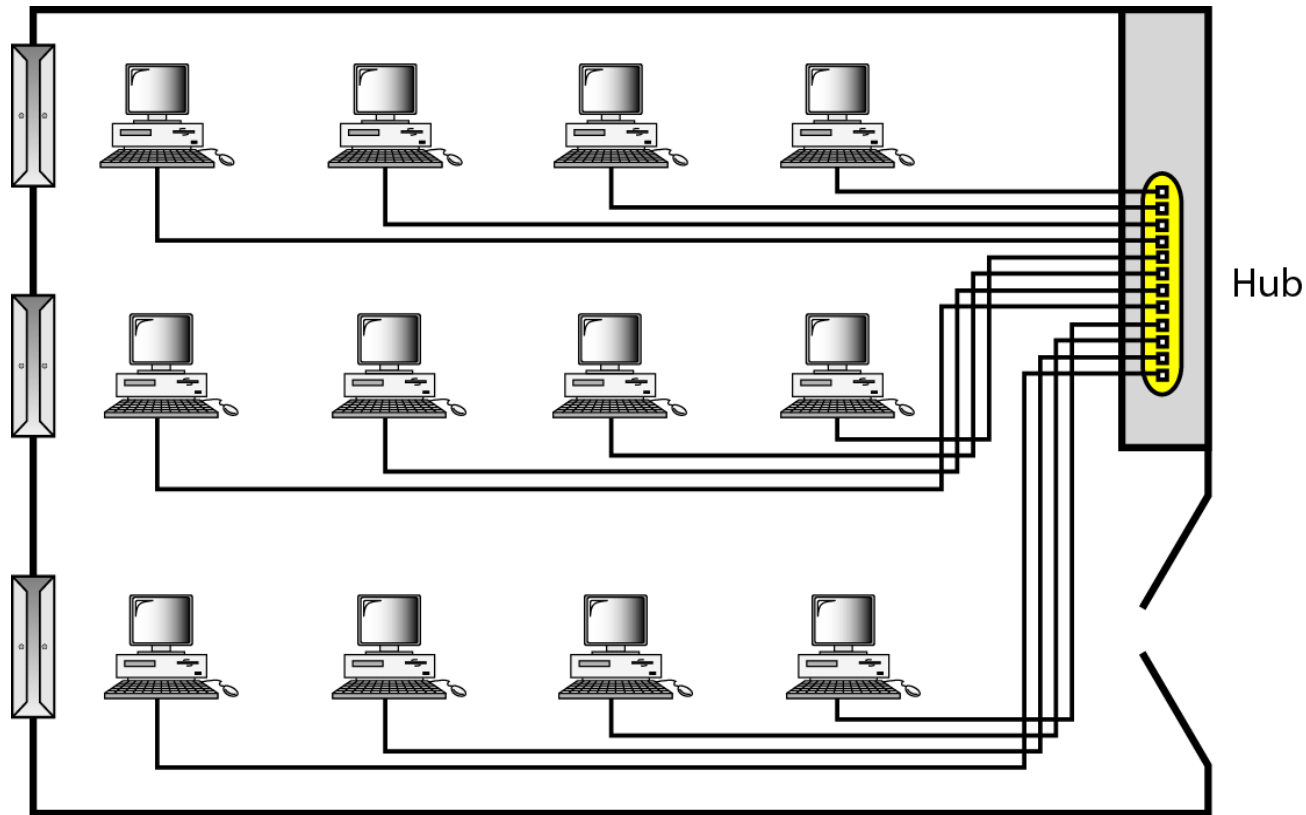
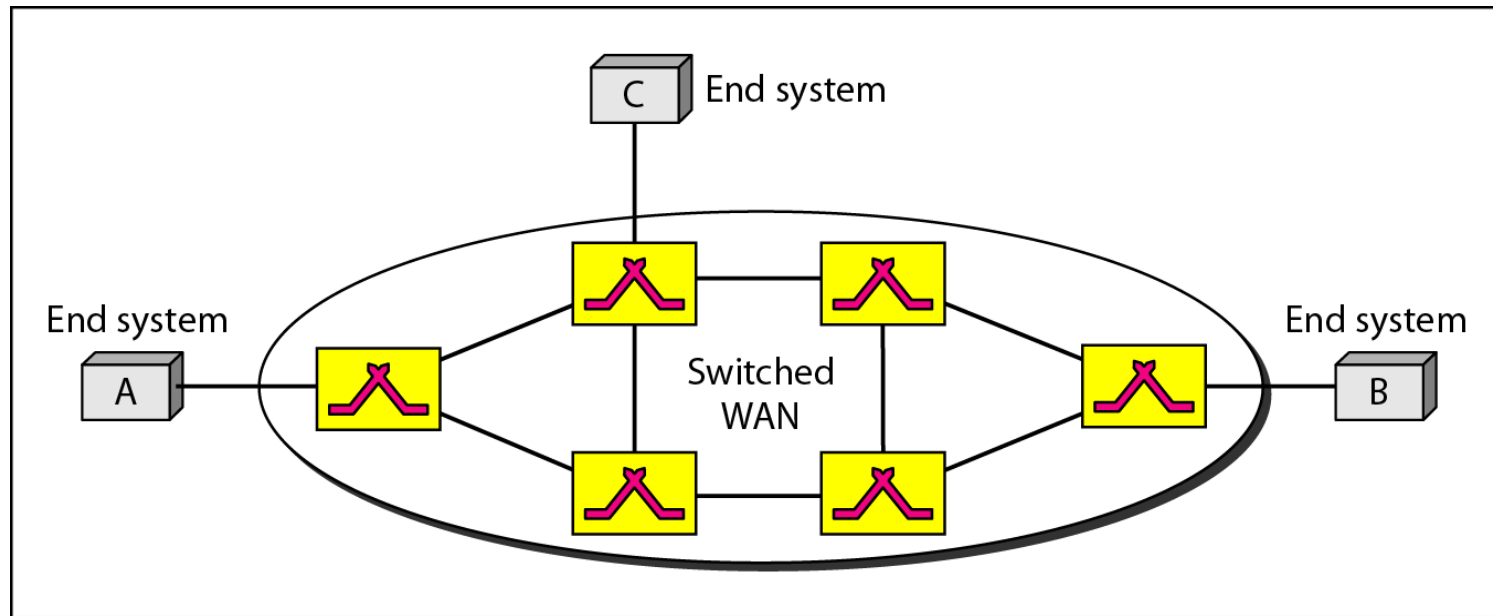
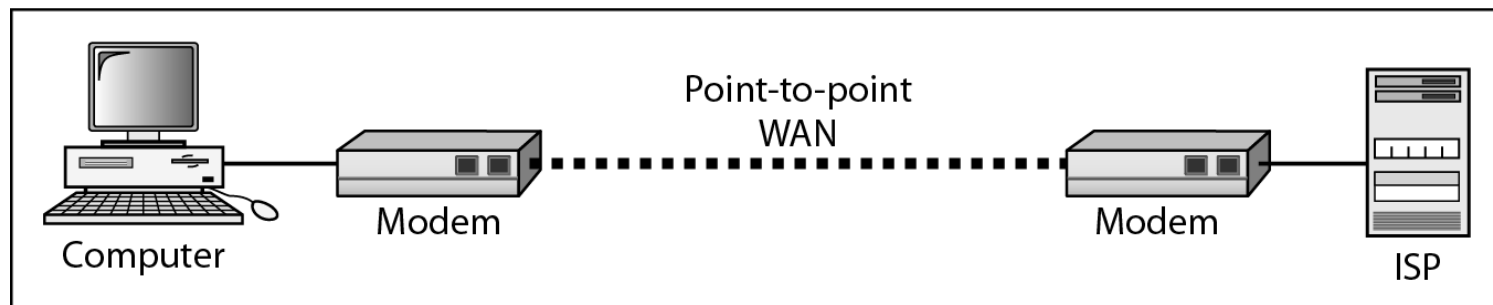


Figure 1.11 *WANs: a switched WAN and a point-to-point WAN*

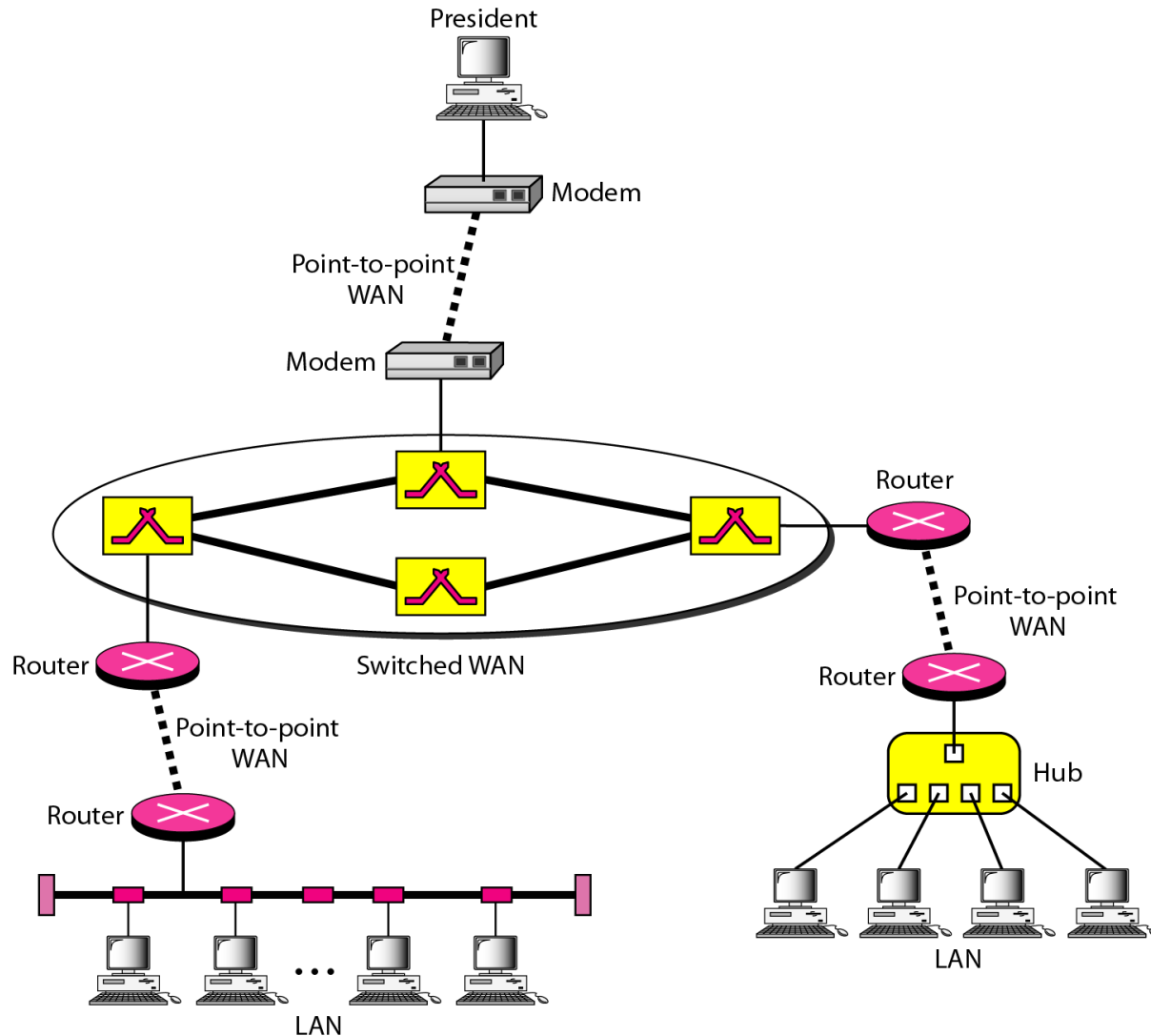


a. Switched WAN



b. Point-to-point WAN

Figure 1.12 *A heterogeneous network made of four WANs and two LANs*



1-4 PROTOCOLS

A protocol is synonymous with rule. It consists of a set of rules that govern data communications. It determines what is communicated, how it is communicated and when it is communicated. The key elements of a protocol are syntax, semantics and timing

Topics discussed in this section:

- Syntax
- Semantics
- Timing

Elements of a Protocol

- **Syntax**
 - Structure or format of the data
 - Indicates how to read the bits - field delineation(design)
- **Semantics**
 - Interprets the meaning of the bits
 - Knows which fields define what action
- **Timing**
 - When data should be sent and what
 - Speed at which data should be sent or speed at which it is being received.

End of Lecture



Thank You